

ZHUOYAN LUO

✉ luoz23@mails.tsinghua.edu.cn · ☎ (+86) 15118658638 · ◇ robertluo1.github.io

EDUCATION

Tsinghua University Shenzhen International Graduate School 2023 – Present

Master student in Artificial Intelligence, expected June 2026

Southeast University, Computer Science and Engineering Department 2019 – 2023

B.S. in Artificial Intelligence

GPA: 4.08/4.8 Ranking: 5/91

EXPERIENCE

BAAI Vision 2025.02 – Present

Research Intern in EMU Team Adviser: Xinlong Wang

Brief introduction: Core contributor in designing a new general visual tokenizer for EMU model.

Project Repository: **Emu3.5** ([Github Stars 1k+](#))

Tencent ARC Laboratory 2024.01 – 2025.02

Research Intern Adviser: Yixiao Ge

Brief introduction: Core contributor in the Visual Tokenizer training and AutoRegressive Image Generation for the Multi-Modal Large Language Model.

Project Repository: **SEED-Voken** ([Github Stars 900+](#))

PUBLICATIONS

- Emu3.5 Team (The Core Contributor). Emu3.5: Native Multimodal Models are World Learners. *Technical Report* In Arxiv, 2025.
- Yong Liu*, **Zhuoyan Luo***, Yicheng Xiao*, Yitong Wang, Shuyan Li, Xiu Li, Yujiu Yang and Yansong Tang. Semantic-Assisted Object Clustering for Multi-Modal Referring Video Segmentation. In *TPAMI*, 2025.
- Wenxuan Wang, Fan Zhang, Yufeng Cui, Haiwen Diao, **Zhuoyan Luo**, Huchuan Lu, Jing Liu, Xinlong Wang. End-to-End Vision Tokenizer Tuning. In *NeurIPS*, 2025.
- Fengyuan Shi*, **Zhuoyan Luo***, Yixiao Ge, Yujiu Yang, Ying Shan, Limin Wang. Scalable Image Tokenization with Index Backpropagation Quantization. In *ICCV*, 2025.
- **Zhuoyan Luo**, Fengyuan Shi, Yixiao Ge, Yujiu Yang, Limin Wang, Ying Shan. OPEN-MAGVIT2: An Open-source Project Toward Democratizing Auto-Regressive Visual Generation. In *Arxiv*, 2024.
- **Zhuoyan Luo**, Yinghao Wu, Yong Liu, Yicheng Xiao, Xiao-Ping Zhang, Yujiu Yang. CoHD: A Counting-Aware Hierarchical Decoding Framework for Generalized Referring Expression Segmentation. In *ICCV*, 2025.
- Yicheng Xiao*, **Zhuoyan Luo***, Yong Liu, Yue Ma, Hengwei Bian, Yatai Ji, Yujiu Yang, Xiu Li. Bridging the Gap: A Unified Video Comprehension Framework for Moment Retrieval and Highlight Detection. In *CVPR*, 2024.
- **Zhuoyan Luo**, Yicheng Xiao, Yong Liu, Shuyan Li, Yitong Wang, Yansong Tang, Xiu Li, Yujiu Yang. SOC: Semantic-Assisted Object Cluster for Referring Video Object Segmentation. In *NeurIPS*, 2023.
- **Zhuoyan Luo**, Yicheng Xiao, Yong Liu, Yitong Wang, Yansong Tang, Xiu Li, Yujiu Yang. 1st Place Solution for Referring Video Object Segmentation. In *ICCV Workshop*, 2023.
- **Zhuoyan Luo**, Yinghao Wu, Shipeng Zhu, Qi Jing, Zihang Xu, Hui Xue. 2nd Place Solution for Multi-line Handwritten Mathematical Expression Recognition Competition. In *ICDAR Workshop*, 2023.
- Qipeng Zhu*, **Zhuoyan Luo***, Shipeng Zhu*, Qi Jing, Zihang Xu, Hui Xue. FATE: A Three-Stage Method for Arithmetical Exercise Correction. In the Journal of *Neural Computing and Applications*, 2023.
- **Zhuoyan Luo**, Hui Xue. Deep Neural Network Based Mathematical Exercises Recognition and Automatic Correction Algorithm. China, 202210783495.4.

AWARDS

1st Prize Award on ICCV 2023 The 5th Large-scale Video Object Segmentation Challenge	Oct. 2023
2nd Prize Award on ICDAR 2023 Multi-line Handwritten Mathematical Expression Recognition	Mar. 2023
2nd Prize Award on iFLYTEK A.I. Handwritten Mathematical Expression Recognition	Nov. 2022
Bronze Prize Award on XIB Competition	May 2024
National Scholarship	2021-2022
Principal Scholarship	2022-2023

PERSONAL

- Research Interest: Multi-Modal Understanding & Generation
- Great Teamwork and Compressive Ability